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CSC 411 Research Methods

Lecture 1: Introduction to Research and the Research Process

Learning Objectives

By the end of this lecture, students should be able to:

1. Understand the concept, purpose, and importance of research.
2. Identify different types and classifications of research.
3. Learn and explain the stages involved in the research process.
4. Recognize ethical considerations in conducting research, including plagiarism.

1. Definition and Purpose of Research

1.1 Meaning of Research

Research is a process of systematic inquiry that entails the collection of data; documentation of critical information; and analysis and interpretation of that data/information, in accordance with suitable methodologies set by specific professional fields and academic disciplines.

Research is conducted to evaluate the validity of a hypothesis or an interpretive framework; to assemble a body of substantive knowledge and findings for sharing them in appropriate manners; and to generate questions for further inquiries.

The term *research* comes from the French word *rechercher*, meaning “to search again.” Research is a **systematic and objective investigation** of a subject or problem to discover facts, establish new knowledge, or validate existing theories. It involves the collection, analysis, and interpretation of data to answer questions or solve problems.

Definition

“Research is a systematic, controlled, empirical, and critical investigation of hypothetical propositions about the presumed relations among natural phenomena.” (Kerlinger, 1986).

Research can be defined as “an activity that involves finding out, in a more or less systematic way, things you did not know” (Walliman and Walliman, 2011, p.7).

“Methodology is the philosophical framework within which the research is conducted or the foundation upon which the research is based” (Brown, 2006).

The Research Methodology chapter of a research describes the research methods, approaches, and designs in detail, highlighting those used throughout the study, justifying my choice by describing the advantages and disadvantages of each approach and design, taking into account their practical applicability to our research.

O’Leary (2004, p.85) describes methodology as the framework that is associated with a particular set of paradigmatic assumptions that we will use to conduct our research.

Allan and Randy (2005) insist that when conducting research methodology should meet the following two criteria:

Firstly, the methodology should be the most appropriate to achieve objectives of the research. Secondly, it should be made possible to replicate the methodology used in other researches of the same nature

The differences between objectivist and subjectivist dimensions are presented by Cohen et al (2007) as taken from Greenfield (1975) in the following manner:

Dimensions of comparison	Objectivist	Subjectivist
Philosophical basis	Realism – the world exists and can be studied as it is	Idealism – the world exists, but is studied differently by different people
Role of social science	Exploring universal laws of the society and the behaviour of people within it	Exploring how the world is interpreted by different people
Basic units of social reality	Society or organisation(s)	Individuals
Comprehension methods	Studying the type and nature of various relationships that allow the collectivity to exist	Studying subjective meanings that individuals impose upon their actions
Theory	A rational construction that has been proposed by researchers in order to explain the human behaviour	Sets of meanings used by individuals in order to interpret their world and behaviour
Research	Validation of theory through experimentation or quasi-experimentation	Looking for meaningful relationships and establishing the consequences of actions
Methodology	The use of quantitative analysis and mathematical methods	The analysis and interpretation of reality
Society	Is managed by a set of general values, rules and regulations	Is managed on the basis of values possessed by people with power

The most comprehensive information regarding vital aspects of methodology is provided by Jackson (2011), which can be summarised in the following table:

Goal met	Research methods	Advantages/disadvantages
Description	Observational method Case study method Survey method	Allows description of behaviour(s) Does not support reliable predictions Does not support cause-and-effect explanations
Prediction	Correlational method Quasi-experimental method	Allows description of behaviour(s) Supports reliable predictions from one variable to another Does not support cause-and-effect explanations
Explanation	Experimental method	Allows description of behaviour(s) Supports reliable predictions from one variable to another Supports cause-and-effect explanations

1.2 The research Philosophy

A research philosophy is a belief about the way in which data about a phenomenon should be gathered, analysed, and used. The term epistemology (what is known to be true), as opposed to doxology (what is believed to be true), encompasses the various philosophies of research approach.

Research philosophy is a vast topic and here we will not be discussing this topic in great details. In business and economics dissertations at Bachelor's level, you are not expected to discuss research philosophy in a great level of depth, and about one page in methodology chapter devoted to research philosophy usually suffices. For a business dissertation at Master's level you may need to provide more discussion of the philosophy of your study, but even there, about two pages of discussion have to be accepted as sufficient by your supervisor.

Discussion of research philosophy in your dissertation should include the following:

1. You need to specify the research philosophy of your study. Your research philosophy can be pragmatism, positivism, realism or interpretivism as discussed below.
2. The reasons behind philosophical classifications of the study need to be provided.
3. You need to discuss the implications of your research philosophy on the research strategy in general and the choice of primary data collection methods in particular.

1.2.1 The Essence of Research Philosophy

Research philosophy deals with the source, nature, and development of knowledge. In simple terms, a research philosophy is a belief about the ways in which data about a phenomenon should be collected, analysed, and used.

Although the idea of knowledge creation may appear to be profound, you are engaged in knowledge creation as part of completing your dissertation. You will collect secondary and primary data and engage in data analysis to answer the research question, and this answer marks the creation of new knowledge.

In essence, addressing research philosophy in your dissertation involves being aware of and formulating your beliefs and assumptions. As it is illustrated in the figure below, the identification of the research philosophy is positioned at the outer layer of the 'research onion'; accordingly, it is the first topic to be clarified in the research methodology chapter of your dissertation.

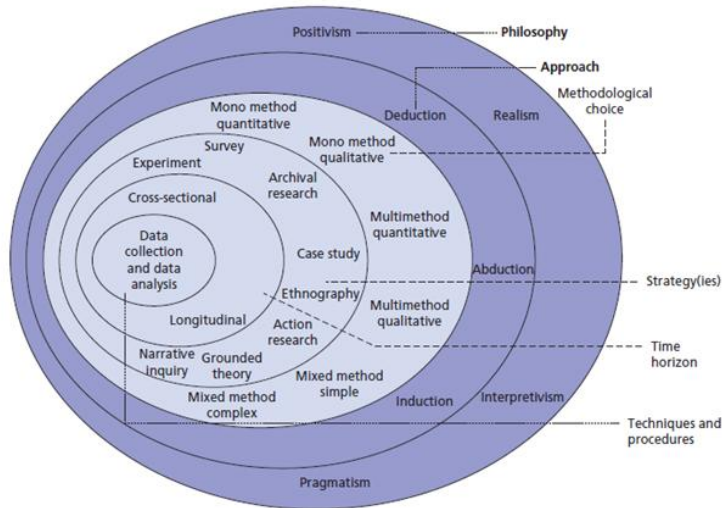


Figure 1. Research philosophy in the ‘research onion

Each stage of the research process is based on assumptions about the sources and the nature of knowledge. The research philosophy will reflect the author’s important assumptions, and these assumptions serve as the base for the research strategy. Generally, research philosophy has many branches related to a wide range of disciplines. Within the scope of business studies in particular, there are four main research philosophies:

1. Pragmatism
2. Positivism
3. Realism
4. Interpretivism (Interpretivist)

1.2.2 The Choice of Research Philosophy

The choice of a specific research philosophy is impacted by practical implications. There are important philosophical differences between studies that focus on facts and numbers such as an analysis of the impact of foreign direct investment on the level of GDP growth and qualitative studies such as an analysis of leadership style on employee motivation in organizations.

The choice between positivist and interpretivist research philosophies or between quantitative and qualitative research methods has traditionally represented a major point of debate. However, the latest developments in the practice of conducting studies have increased the popularity of pragmatism and realism philosophies as well.

Moreover, as it is illustrated in table below, there are popular data collection methods associated with each research philosophy.

	Pragmatism	Positivism	Realism	Interpretivism
Popular data collection method	Mixed or multiple method designs, quantitative and qualitative	Highly structured, large samples, measurement, quantitative, but can use qualitative	Methods chosen must fit the subject matter, quantitative or qualitative	Small samples, in-depth investigations, qualitative

Figure 2. Research philosophies and data collection methods

1.3 Objectives of Research

Research serves several objectives, including:

- **Exploration:** To gain familiarity with a phenomenon or identify new insights.
- **Description:** To describe characteristics or functions of phenomena.
- **Explanation:** To explain why certain phenomena occur.
- **Prediction:** To forecast future occurrences or behaviors.
- **Application:** To apply knowledge in solving practical problems.

1.4 Characteristics of Scientific Research

1. **Systematic:** Follows an organized and logical sequence of steps.
2. **Empirical:** Based on observable and measurable evidence.
3. **Objective:** Free from personal bias or subjectivity.
4. **Replicable:** Can be repeated by others to verify findings.

5. Logical: Each step is justified by reasoning and evidence.
6. Cumulative: Builds upon existing knowledge.

2. Types of Research

2.1 Based on Purpose

1. Basic (Pure) Research:

- Aimed at expanding knowledge without immediate practical application.
- Example: Studying the structure of atoms or linguistic patterns in African languages.

2. Applied Research:

- Focuses on solving real-world problems or improving practices.
- Example: Developing an AI model for mental health screening.

2.2 Based on Methodology

1. Qualitative Research:

- Focuses on understanding human behavior, experiences, or social phenomena.
- Uses interviews, observations, case studies, etc.
- Data are non-numerical and interpretive in nature.

2. Quantitative Research:

- Emphasizes measurement, statistical analysis, and numerical data.
- Uses surveys, experiments, or statistical models.
- Example: Measuring the impact of a new teaching method on student performance.

2.3 Based on Nature or Design

1. Descriptive Research:

- Describes characteristics or functions of a situation or group.
- Example: Surveying student attitudes toward online learning.

2. Exploratory Research:

- Conducted when the problem is not clearly defined.

- Seeks to gain insights and refine research questions.

3. Explanatory (Causal) Research:

- Seeks to explain relationships between variables.
- Example: Investigating whether stress causes poor academic performance.

4. Experimental Research:

- Involves manipulation of variables to determine cause-and-effect relationships.
- Example: Testing a new drug's effect on blood pressure.

3. The Research Process

The research process provides a roadmap for conducting scientific inquiry. It generally includes the following stages:

1. Identification of Research Problem

- Recognize an issue or knowledge gap needing investigation.
- A good research problem should be specific, researchable, and significant.

2. Review of Related Literature

- Study existing works to understand the current state of knowledge.
- Helps refine the research problem and identify suitable methods.

3. Formulation of Research Questions or Hypotheses

- **Research Questions:** Direct queries the study seeks to answer.
- **Hypotheses:** Testable predictions about relationships between variables.

4. Research Design and Methodology

- Decide on the type of research (qualitative/quantitative/mixed).
- Choose data collection instruments and sampling methods.

5. Data Collection

- Gather relevant data through surveys, interviews, experiments, or observations.

6. Data Analysis and Interpretation

- Organize, process, and analyze data using appropriate tools (e.g., SPSS, Excel).
- Interpret results to answer the research questions or test hypotheses.

7. Drawing Conclusions and Recommendations

- Summarize findings and discuss implications or applications of the results.

8. Report Writing and Presentation

- Prepare a structured research report or thesis following academic standards.
- Include sections such as Introduction, Methodology, Results, Discussion, and Conclusion.

4. Ethics in Research

4.1 Ethical Considerations

Ethics are moral principles that guide the researcher's conduct. Key considerations include:

- **Informed Consent:** Participants must willingly agree to be part of the research.
- **Confidentiality:** Protect participants' privacy and data.
- **Integrity:** Report data and findings honestly, without manipulation.
- **Non-maleficence:** Avoid causing harm to participants or communities.
- **Transparency:** Acknowledge funding sources and avoid conflicts of interest.

4.2 Plagiarism

- Plagiarism is the act of presenting someone else's ideas or words as one's own without proper acknowledgment.
- It is an academic offense and violates research integrity.
- **Ways to Avoid Plagiarism:**
 - Cite all sources accurately using recognized citation styles (APA, MLA, etc.).
 - Paraphrase appropriately without copying sentences.
 - Use plagiarism detection tools to ensure originality.

Summary

Research is a structured and scientific process aimed at discovering new knowledge or solving problems. It involves identifying a problem, reviewing literature, collecting and analyzing data,

and presenting findings. Understanding the research process and adhering to ethical standards ensures credibility, reliability, and societal relevance.